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                   --- Welcome to Maude ---
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           Maude 3.5.1 built: Jul 16 2025 12:00:00
            Copyright 1997-2025 SRI International
                  Tue Apr 28 11:29:09 2026
```

reduce in ST : lambda "x" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" "x"

reduce in ST : lambda "x" lambda "y" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" lambda "y" "x"

reduce in ST : app lambda "x" app "x" "x" lambda "x" app "x" "x" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: app lambda "x" app "x" "x" lambda "x" app "x" "x"

reduce in ST : lambda "x" app lambda "y" "y" lambda "z" "z" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: lambda "x" app lambda "y" "y" lambda "z" "z"

reduce in ST : app app lambda "x" lambda "y" app app "x" "y" "x" lambda "x" lambda "y" "x" lambda "x" lambda "y" "y" .
rewrites: 0 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result Term: app app lambda "x" lambda "y" app app "x" "y" "x" lambda "x" lambda "y" "x" lambda "x" lambda "y" "y"

reduce in ST : *inference*(lambda "x" app "x" "x") .
rewrites: 61 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *error*

reduce in ST : *inference*(lambda "x" lambda "y" "x") .
rewrites: 20 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(2) -> t(1))

reduce in ST : *inference*(lambda "x" lambda "y" "y") .
rewrites: 18 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(2) -> t(1) -> t(1))

reduce in ST : *inference*(lambda "x" "z") .
rewrites: 11 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*("z" : t(1), t(2) -> t(1))

reduce in ST : *inference*(lambda "x" app lambda "y" "y" lambda "z" "z") .
rewrites: 103 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(4) -> t(2) -> t(2))

reduce in ST : *inference*(lambda "x" "x") .
rewrites: 12 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(1))

reduce in ST : *inference*(lambda "x" lambda "y" "x") .
rewrites: 20 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(1) -> t(2) -> t(1))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z") .
rewrites: 239 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(5) -> t(6) -> t(7)) -> (t(5) -> t(6)) -> t(5) -> t(7))

reduce in ST : *inference*(app lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z" lambda "a" "a") .
rewrites: 451 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(6) -> t(7)) -> t(6)) -> (t(6) -> t(7)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" "y") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(4) -> t(2) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" app lambda "z" "x" app "y" "x") .
rewrites: 208 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(7) -> (t(7) -> t(6)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" app app "y" lambda "z" app app "z" "x" "x" "x") .
rewrites: 378 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(8) -> (((t(8) -> t(8) -> t(6)) -> t(6)) -> t(8) -> t(9)) -> t(9))

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" lambda "z" app "y" "z") .
rewrites: 255 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(5) -> t(6)) -> (t(5) -> t(6)) -> t(7)) -> (t(5) -> t(6)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" "y") .
rewrites: 165 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(4) -> t(4) -> t(5)) -> t(4) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app "x" app "y" "z") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(4) -> t(5)) -> (t(3) -> t(4)) -> t(3) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "y" "z") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(2) -> t(4) -> t(5))

reduce in ST : *inference*(lambda "x" lambda "y" lambda "z" app app "x" "z" "y") .
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, (t(2) -> t(4) -> t(5)) -> t(4) -> t(2) -> t(5))

It should have been "x" and not "z" in the end:

lambda "x" lambda "y" app lambda "z" "x" app "y" "x"

result is: p(nil, t(7) -> (t(7) -> t(6)) -> t(7))

reduce in ST : *inference*(lambda "x" lambda "y" app lambda "z" "x" app "y" "z") .
rewrites: 166 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*("z" : t(5), t(7) -> (t(5) -> t(6)) -> t(7))

---

reduce in ST : *inference*(lambda "x" lambda "y" app app "y" lambda "z" app app "z" "x" "x" "x") .
rewrites: 378 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, t(8) -> (((t(8) -> t(8) -> t(6)) -> t(6)) -> t(8) -> t(9)) -> t(9))

reduce in ST : *inference*(lambda "x" lambda "y" app app "x" "y" lambda "z" app "y" "z") .
rewrites: 255 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result PPair: *p*(*nil*, ((t(5) -> t(6)) -> (t(5) -> t(6)) -> t(7)) -> (t(5) -> t(6)) -> t(7))

reduce in POLY : *w*(*nil*, let "i" = lambda "x" "x" in "i", 1) .
rewrites: 37 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: *wpair*(*nil*, t(2) -> t(2), 3)

reduce in POLY : *w*(*nil*, lambda "x" let "y" = "x" in "y", 1) .
rewrites: 32 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: *wpair*(*nil*, t(1) -> t(1), 2)

reduce in POLY : *w*(*nil*, let "i" = lambda "x" "x" in app "i" "i", 1) .
rewrites: 113 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: *wpair*(t(4) = t(3) -> t(3) t(2) = t(3) -> t(3), t(3) -> t(3), 5)

reduce in POLY : $w(nil,$ let "i" = lambda "x" let "y" = "x" in "y" in app "i" "i", $1)$.
rewrites: 132 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(4) = t(3) -> t(3) t(2) = t(3) -> t(3), t(3) -> t(3), $5)$

reduce in POLY : $w(nil,$ lambda "m" let "y" = "m" in let "x" = app "y" lambda "x" lambda "y" "x" in "x", $1)$.
rewrites: 155 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(1) = (t(2) -> t(3) -> t(2)) -> t(4), ((t(2) -> t(3) -> t(2)) -> t(4)) -> t(4), $5)$

reduce in POLY : $w(nil,$ let "k" = lambda "z" lambda "w" app "z" "w" in lambda "z" app "z" app "k" "k", $1)$.
rewrites: 442 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(1) = t(2) -> t(3) t(9) = (t(7) -> t(8)) -> t(7) -> t(8) t(6) = t(7) -> t(8) t(5) = t(7) -> t(8) t(4) = ((t(7) -> t(8)) -> t(7) -> t(8)) -> t(10), (((t(7) -> t(8)) -> t(7) -> t(8)) -> t(10)) -> t(10), $11)$

reduce in POLY : $w(nil,$ let "k" = lambda "x" lambda "y" "x" in let "w" = lambda "x" lambda "y" lambda "z" app app "x" "y" "z" in app app "k" "w" "w", $1)$.
rewrites: 946 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(3) = t(4) -> t(6) t(6) = t(5) -> t(7) t(13) = t(9) -> (t(10) -> t(11) -> t(12)) -> t(10) -> t(11) -> t(12) t(8) = (t(10) -> t(11) -> t(12)) -> t(10) -> t(11) -> t(12) t(17) = (t(10) -> t(11) -> t(12)) -> t(10) -> t(11) -> t(12) t(9) = (t(14) -> t(15) -> t(16)) -> t(14) -> t(15) -> t(16), (t(10) -> t(11) -> t(12)) -> t(10) -> t(11) -> t(12), $18)$

reduce in POLY : $w(nil,$ let "s" = lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z" in let "k" = lambda "x" lambda "y" "x" in let "i" = lambda "x" "x" in app app "s" "k" "i", $1)$.
rewrites: 1165 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(1) = t(3) -> t(4) t(2) = t(3) -> t(5) t(4) = t(5) -> t(6) t(15) = (t(13) -> t(14)) -> t(13) -> t(13) t(12) = t(13) t(11) = t(14) t(10) = t(13) t(17) = t(16) -> t(16) t(14) = t(16) t(13) = t(16), t(16) -> t(16), $18)$

reduce in POLY : $w(nil,$ app let "s" = lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z" in let "k" = lambda "x" lambda "y" "x" in app "s" "k" lambda "x" lambda "y" lambda "z" app app "x" "z" app "y" "z", $1)$.
rewrites: 1330 **in** 0 **ms cpu** (0 **ms real**) (~ **rewrites/second**)
result WPair: $wpair($t(1) = t(3) -> t(4) t(2) = t(3) -> t(5) t(4) = t(5) -> t(6) t(14) = (t(12) -> t(13)) -> t(12) -> t(12) t(11) = t(12) t(10) = t(13) t(9) = t(12) t(15) = t(17) -> t(18) t(16) = t(17) -> t(19) t(18) = t(19) -> t(20) t(21) = (t(17) -> t(19) -> t(20)) -> t(17) -> t(19) -> t(20) t(13) = (t(17) -> t(19)) -> t(17) -> t(20) t(12) = t(17) -> t(19) -> t(20), (t(17) -> t(19) -> t(20)) -> t(17) -> t(19) -> t(20), $22)$

Bye.